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Roll No

IT-4002-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Database Management System

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain in detail about Database Management System advantages over file management system. 7
b) Explain the importance of Database models. Also compare relational data model and Network Data model. 7
2. a) Explain what is meant by one-to-one, one-to-many and many-to-many relationships between entities, giving an example of each. 7
b) Design an E-R diagram for keeping track of the exploits of your favourite sports team. You should store the matches played, the scores in each match, the players in each match and individual player statistics for each match. Summary statistics should be modeled as derived attributes. 7
3. a) Consider the following schema for a library database
Author (Authurname, citizenship, birthyear)
Book(isbn, title, authurname)
Topic(isbn, subject)
Branch(libname, city)

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Instock(isbn, libname, quantity)

Write relational algebra expressions for the following queries 7

- i. Give all authors born after 1940.
- ii. Give the names of libraries in Sydney.
- iii. Give the cities where each book is held.
- iv. Give the title of each book on the topic of either alcohol or drugs.
- b) Define the concept of aggregation. Give two examples of where this concept is useful? 7

4. a) Consider the following schema for a Employee database.
Emp(empno, ename, Job, sal, Deptno, hiredate)

Write SQL for the following queries: 7

- i. Display all the information of the emp table.
 - ii. Display unique jobs from EMP table.
 - iii. List the details of the emps in asc order of their salaries.
 - iv. List the details of the emps in asc order of the Deptnos and desc of Jobs.
 - v. Display all the unique job groups in the descending order.
 - b) Explain Query Optimization with suitable example. 7
5. a) Explain briefly about 3NF, 4NF and BCNF with suitable examples. 7
- b) What is Functional Dependency? Explain types and properties of FD's. 7

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6. a) List and explain various issues while transactions are running concurrently in DBMS. 7
b) Explain two phase locking protocol. Also list advantages of this protocol. 7
7. a) What is serializability? Explain view and conflict serializability with example. 7
b) Explain following relational algebra operation: 7
 - i) Natural join operation
 - ii) Selection and projection operation
8. a) Explain the difference between a distributed database and distributed processing. 7
b) Explain different types of keys available in Relational Model. 7
